XIAMEN UNIVERSITY MALAYSIA

ONLINE FINAL EXAMINATION

Course Code:	CST103			
Course Name:	Programming Language C			
Question Paper Setter:	Hamid Tahaei			
Academic Session:	2020/09	Question Paper:	A ☑	B ☐
Total No. of Pages:	2	Time Allocated:	2 hours	
Additional Materials:	-			
Apparatus Allowed:	-			

INSTRUCTIONS TO CANDIDATES

1. This paper consists of 1 Section(s):

   Section A: 3 question(s)

   Answer all questions.

2. Please follow the requirement of each section and type in all the corresponding answers on the downloadable answer book prosvided.

3. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Section A (100 marks)

Read Code

Question 1 (25 marks)

The value of f x can be calculated from the infinite series.

$$x = 1 - \frac{1}{2} + \frac{2}{3} - \frac{3}{4} + \frac{4}{5} - \frac{5}{6} + \frac{6}{7}$$

Write a program to input the number of terms and calculate the corresponding value of x according to the above formula. Examples of result are shown in Figure 1, 2 and 3. Please solve the problem with function using pass-by-reference, and follow the input and output format of the results as shown below.

```
Enter the number of temrs: 4
The result is: 1.216667
```

Figure 1

```
Enter the number of temrs: 2
The result is: 1.166667
```

Figure 2

```
Enter the number of temrs: 3
The result is: 0.416667
```

Figure 3

Question 2 (25 marks)

Write a program to input some characters from keyboard until the input character is newline "\n", and count the number of letters and the number of digits in the string. Examples of result are as following. Please solve the problem with array(s), and please follow the input and output format of the results as shown below.

```
Please input some characters
;'123WFEFabc234
The number of digits is 6
The number of letters is 7
```

```
Please input some characters
qw122ABC';<
The number of digits is 3
The number of letters is 5
```

Question 3 (50 marks)

Suppose there is only one student and this student has 4 courses. Please write a program to input the student's information with names and marks of 4 courses, and calculate the total marks of 4 courses for this student. Please follow the format shown in the following figure, and please make sure to use all the following methods in one program.

1) Array(s)
2) Structure(s)
3) Function(s)
4) Pointer(s)

```
Enter Number1 student's information:
Enter name: Peter
Enter Math Mark: 2
Enter Science Mark: 3
Enter Art Mark: 4

Enter Number2 student's information:
Enter name: Linda
Enter Math Mark: 3
Enter Science Mark: 4
Enter Art Mark: 5

Enter Number3 student's information:
Enter name: Mike
Enter Math Mark: 2
Enter Science Mark: 4
Enter Art Mark: 5

Enter Number4 student's information:
Enter name: Lisa
Enter Math Mark: 3
Enter Science Mark: 4
Enter Art Mark: 4

The total marks of 3 courses for each student are:
Peter's total mark is: 9
Linda's total mark is: 12
Mike's total mark is: 11
Lisa's total mark is: 11

The average marks of 4 students for each course are:
The average Math mark of 4 students is: 2.500000
The average Science mark of 4 students is: 3.750000
The average Art mark of 4 students is: 4.500000
```

- END OF PAPER -